

Article

Tilt-Rotor Tricopter with State-Constrained Controller Design

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Highlights

What are the main findings?

- A state-constrained backstepping control architecture integrated with a barrier Lyapunov function (BLF), a first-order filter, and a linear extended state observer (LESO) is proposed, effectively addressing state constraints, the “complexity explosion” issue in backstepping, and internal/external disturbances during tilt-rotor tricopter transition.
- An improved Aquila Optimizer (AO) with a chaotic elite opposition-based learning initialization strategy is developed to optimize controller hyperparameters, achieving superior steady-state and transient control performance compared to traditional backstepping and PID controllers.

What are the implications of the main findings?

- The proposed control framework provides a systematic and practical solution for state-constrained transition control of tilt-rotor UAVs, offering enhanced robustness and reliability for complex flight scenarios involving mode switching.
- The integration of metaheuristic optimization with state-constrained backstepping control sets a precedent for automated parameter tuning in nonlinear UAV control systems, reducing reliance on manual tuning and promoting broader applicability to other underactuated/overactuated aerial vehicles.

Abstract

This paper presents a control architecture based on Pixhawk4 to address the transition mode control issue of a tilt-rotor tricopter. While the classical PID controller designed for the rotor mode can handle most engineering applications under normal environmental conditions, it does not fully consider disturbances such as those arising from internal perturbations or the external environment. In response, this paper proposes a controller design with disturbance observation to meet the robustness requirements of the unmanned aerial vehicle (UAV) under perturbed conditions. For the state-constrained control problem of the UAV, the original model is simplified, and a three-channel model for control purposes is introduced. By incorporating a barrier Lyapunov function, the state constraint problem of the UAV is solved, ensuring a smooth transition from rotor mode to fixed-wing mode. To address the high complexity of virtual control law derivatives in backstepping control, a filter is used to estimate the derivatives of the UAV's virtual states, and a linear extended state observer is introduced to estimate external and transitional disturbances during flight. Lyapunov theory is employed to ensure the closed-loop stability of the control scheme. Finally, inspired by the hunting process of aquilas, the Aquila Optimizer (AO) optimization algorithm is applied to optimize the controller's hyperparameters, further enhancing the reliability and transient performance of the control system.



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1. Introduction

Tilt-rotor tricopter drones face several challenges during transition mode, including coupled control channels, system nonlinearities, and partial underactuation or overactuation. The tilting motion of the rotors significantly impacts attitude, airspeed, and stability during the transition phase. Therefore, designing an appropriate transition strategy, along with stable controllers and switching mechanisms, is crucial for achieving smooth and stable transitions. Common control methods for tilt-rotor drones include gain scheduling, adaptive control, dynamic inversion, and backstepping. Additionally, sliding mode control, widely used in aircraft control, is also considered for tilt-rotor drones due to its applicability in multi-mode control switching.

Gain scheduling involves applying linear and classical control theory to linearized operating points, designing corresponding linear controllers for each point, and then scheduling the control gains based on the operating conditions. While this method offers high reliability and operability, it suffers from complexity and poor robustness, making it difficult to suppress disturbances caused by uncertainties under complex conditions [1]. Several researchers have applied this method to tilt-rotor tricopter transition control. For instance, J. Holsten et al. employed PID control for transition mode, optimizing controller gains through a multi-objective parameter synthesis approach. Their simulation results met predefined performance criteria [2]. Lior Zivan et al. designed a cascaded PID control system for the “Panther” tilt-rotor drone, with the outer loop controlling altitude via throttle and airspeed via pitch, while the inner loop stabilized attitude using angular rate control. Flight tests showed transition times under 3 s, with quick recovery from initial attitude deviations [3]. Reference [4] designed a miniature tilt-tricopter UAV, for which the researchers developed a flight controller employing altitude hold control for the vertical channel and a PI controller for attitude and angular rate regulation. Simulations demonstrated that the entire transition process completed in approximately 10 s under disturbance-free conditions while maintaining stable attitude. Kissoum et al. proposed a robust fractional-order adaptive gain-scheduled control strategy, integrating linear parameter-varying (LPV) models with fractional-order PID controllers and supervising a set of pre-tuned linear quadratic regulators (LQR) through an adaptive switching law, which significantly enhanced the robustness of civil UAVs in airspeed and altitude control, and numerical simulations verified that this method could maintain the stability of the closed-loop system under both continuous and discontinuous parameter variation scenarios [5].

Adaptive control employs online parameter identification to adjust control laws dynamically, ensuring that closed-loop performance meets desired specifications [6]. Yildiray Yildiz et al. applied model-reference adaptive control to a tilt-quadrotor drone, designing altitude and attitude controllers for transition mode. Simulations confirmed stable transitions within 10 s, demonstrating the method’s effectiveness against structural uncertainties [7]. Guo et al. proposed a finite-time adaptive sliding mode fault-tolerant control method, combining fast terminal sliding mode control with a certainty-equivalent adaptive mechanism, the adaptive law real-time estimates composite actuator fault parameters, and the hyperbolic tangent function is used to suppress chattering, ensuring that the system converges to the equilibrium point within a specified time after a fault occurs, which exhibits faster transient response and higher steady-state accuracy compared to traditional methods [8].

He et al. designed an enhanced active disturbance rejection adaptive control strategy based on RBF neural networks, taking the real-time calculated control inputs of the controller and the output states of the UAV as the inputs of the neural network to dynamically adjust the parameters of the extended state observer (ESO) and the nonlinear state error feedback control law (NLSEF), which effectively compensates for model uncertainties and external complex disturbances of tilt-quadrotor UAVs; simulation results show that its anti-disturbance performance and stability are superior to those of the traditional ADRC controller [9].

Dynamic inversion transforms nonlinear systems into linear ones via state feedback, enabling the use of linear control design methods. However, its sensitivity to disturbances and modeling errors necessitates highly accurate models or compensatory techniques like neural networks [10]. Yongjun Seo et al. used quaternion-based dynamic inversion for attitude control in a tilt-tricopter, achieving stable and energy-efficient performance [11]. Xia Qingyuan et al. combined dynamic inverse design with neural network compensation to investigate a tilt-rotor dual-rotor UAV system. They designed a transition mode control system and developed a simulation test environment for the tilt-rotor aircraft [12], validating the effectiveness of the control algorithm. Liu et al. proposed an INDI-based control scheme for quadrotor TRUAVs, establishing a control-oriented longitudinal motion model and designing cascaded inner/outer loop controllers for altitude, velocity, and attitude during transitions. They integrated virtual control allocation and an adaptive tilt strategy to handle aerodynamic coupling between rotors and wings. Simulation results validated that the scheme effectively mitigates transition-induced instability and enhances tracking accuracy of key states [13].

The basic idea of backstepping control is to decompose a high-dimensional nonlinear system into multiple subsystems with orders no higher than that of the original system. For each subsystem, a local Lyapunov function and a virtual controller are designed, recursively working backward until the final subsystem is reached, ultimately yielding a control law that ensures overall system stability [14]. The backstepping design process is systematic and methodical, capable of handling mismatched disturbances while guaranteeing the dynamic performance of each subsystem. However, it also has notable drawbacks, such as difficulties in achieving high-precision control and the issue of “complexity explosion” [15]. Anil Kulhare et al. designed a backstepping controller for a tri-rotor UAV with a tiltable tail rotor, and simulation tests verified that the controller could achieve position tracking and hover control [16]. Additionally, Chowdhury et al. developed a backstepping-based PD controller for a tilt-rotor dual-rotor UAV and conducted simulation experiments, with results demonstrating effective trajectory tracking performance [17]. In recent years, Borja-Jaimes et al. proposed a backstepping sliding mode control framework integrated with a super-twisting observer (STO); relying only on the direct measurement of position and attitude, the STO real-time reconstructs linear and angular velocity information and inputs it into the control law, without the need for full-state feedback or explicit disturbance compensation, and three typical simulation scenarios show that the stability and tracking consistency of the closed-loop system are significantly superior to those of the traditional backstepping control method [18].

Sliding mode control (SMC), known for its robustness against uncertainties, uses a sliding surface to guide system states to equilibrium. A key advantage of this method is its strong robustness against disturbances such as modeling errors and parameter uncertainties [19]. Yao Zou applied SMC to quadrotor and tilt-quadrotor drones, achieving precise trajectory tracking and disturbance rejection [20]. Liu Haibo et al. designed a fractional-order sliding mode controller based on a second-order sliding mode disturbance observer for a tilt-rotor quadcopter UAV, achieving attitude step control in rotor-borne

mode [21]. Zhu Xueping et al. developed an inner-loop sliding mode controller for attitude angle control of a tilt-rotor aircraft model and established a simulation environment. Flight simulation results demonstrated that the controller is suitable for transition mode control of the tilt-rotor model [22]. Zhang et al. proposed an adaptive super-twisting sliding mode control method, constructing a dual-loop control system where both the inner and outer loops adopt a super-twisting sliding mode structure, and an interference adaptive law is introduced to dynamically compensate for the control input; MATLAB simulations verify that this method not only improves trajectory tracking accuracy and robustness but also effectively suppresses chattering under low control gains, with the chattering amplitude reduced by more than 50% compared with the traditional sliding mode control [23].

Refs. [16–18] focus on achieving basic stability and trajectory tracking, but the state constraints, systematic disturbance rejection using observers, or the “complexity explosion” issue inherent in backstepping for higher-order systems are not addressed, and parameter tuning is typically manual. SMC is renowned for its robustness against disturbances and uncertainties. However, a well-known challenge of SMC is chattering, which can excite high-frequency dynamics and is undesirable for smooth actuator operation. Refs. [20–23] leverage the robustness of VSC/SMC but inherits the associated chattering problem, and the design complexity is high. The backstepping method proves particularly effective for managing the cascaded nature of flight dynamics. Its recursive design approach allows a complex n th-order system to be decomposed into a series of n first-order subsystems. This decomposition facilitates the systematic implementation of both adaptive and robust control designs. This study comprehensively considers the requirements of UAV application scenarios and the complexity of control algorithms, adopting the backstepping method as the fundamental control strategy. The main contributions of this paper are listed as follows.

- (1) A state-constrained backstepping control architecture is proposed for the transition mode control of tilt-rotor tricopters. To address the limitations of classical PID controllers (e.g., poor robustness against internal/external disturbances), the design integrates three key components: a barrier Lyapunov function (BLF) to ensure state variables remain within predefined bounds, a first-order filter to mitigate the “complexity explosion” issue inherent in backstepping control (by estimating derivatives of virtual control laws), and a linear extended state observer (LESO) to compensate for flight disturbances (including internal perturbations and external environmental interference).
- (2) A control-oriented model simplification strategy is developed. The original nine-dimensional rigid-body dynamic model of the tilt-rotor tricopter is simplified into a three-channel model (considering roll/pitch/yaw angles and 3D position) by introducing reasonable assumptions (e.g., neglecting small inertia products and gyroscopic effects). This simplification balances simulation accuracy and computational complexity, enabling practical controller implementation while retaining core dynamic characteristics.
- (3) An improved Aquila Optimizer (AO) algorithm is proposed for controller hyperparameter optimization. To overcome the insufficient diversity of initial populations in the traditional AO, a chaotic elite opposition-based learning initialization strategy is introduced. This strategy enhances the global exploration capability and convergence speed of the algorithm, ensuring the optimized controller achieves superior steady-state and transient performance (e.g., faster response and smaller tracking errors).
- (4) A dedicated hardware-in-the-loop (HIL) simulation platform is constructed based on Pixhawk4 and MATLAB Simulink. Leveraging the Pixhawk Pilot Support Package (PSP) Toolbox, the platform realizes automatic compilation and deployment of Simulink-based control algorithms to Pixhawk hardware. This platform effectively

validates the proposed control strategy under realistic flight conditions, verifying its robustness and applicability for tilt-rotor tricopter transition mode control.

2. Tilt-Rotor Tricopter UAV

2.1. UAV Configuration

The tilt-rotor tricopter UAV studied in this paper is illustrated in Figure 1, and related parameters is shown in Table 1. Through multiple optimizations in structural design, the aircraft features a lifting-body configuration that integrates seamlessly with the main wing, ensuring excellent aerodynamic stability and high flight efficiency. The UAV is equipped with three rotors of identical specifications—two mounted at the front and one at the rear. The front rotors are tilt-capable (The tilt angles are denoted by α_1 and α_2 in Figure 1), while the rear rotor has a fixed deflection angle (which can be represented by α_3 in Figure 1). The flight mode of the UAV is determined by the tilt angle of the front rotors, with different tilt angles corresponding to distinct flight modes, which are rotor mode, fixed-wing mode, and transition mode. Depending on takeoff/landing conditions and mission payload requirements, the UAV can achieve both STOL (Short Takeoff and Landing) and VTOL (Vertical Takeoff and Landing) operations.

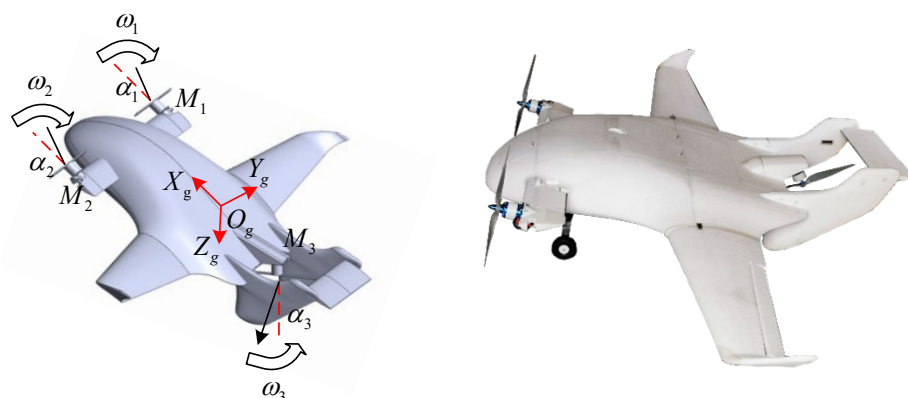


Figure 1. Tilt-rotor tricopter UAV model and physical prototype.

Table 1. Basic parameters of the UAV.

Parameter	Value
Wingspan (mm)	2600
Fuselage Length (mm)	1480
Fuselage Airfoil	GOE777
Wing Airfoil	GOE621
Mean Wing Chord (mm)	221
Wing–Body Area (mm ²)	178,458
Maximum Takeoff Weight (kg)	15
Payload Capacity (kg)	6
Cruise Speed (m/s)	15–25
Mission Altitude (m)	20–500

2.2. Flight Control Architecture

The Pixhawk 4 open-source flight control development kit is adopted as the basic hardware architecture of the UAV flight control system in this paper, as shown in Figure 2. Co-developed by Holybro and Auterion, the Pixhawk 4 features in-depth optimization of the PX4 autopilot and comes pre-installed with native PX4 firmware (px4fmu-v5). Pixhawk 4 adopts the NuttX real-time operating system, and its processor system is composed of a main processor and a coprocessor. The main FMU processor is equipped with an

STM32F765, a 32-bit Arm® Cortex®-M7 core operating at a clock frequency of 216 MHz, with 2 MB of flash memory and 512 KB of RAM, and the IO processor adopts an STM32F100, a 32-bit Arm® Cortex®-M3 core running at 24 MHz, integrated with 8 KB of SRAM. This strong real-time processing capability ensures the reliable execution of complex model-based algorithms and satisfies the performance requirements of the control algorithms proposed in this study.

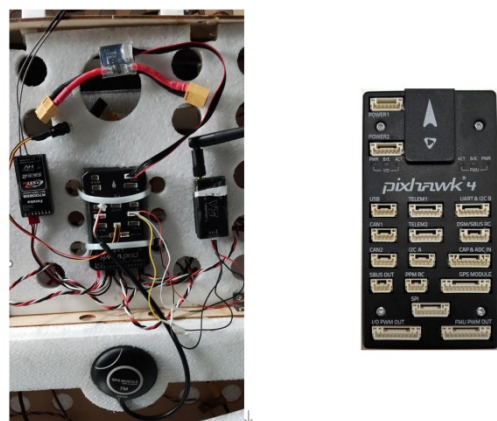


Figure 2. Pixhawk4 flight control hardware of the UAV.

As shown in Figure 3, the Pixhawk 4 open-source flight control development kit serves as the hardware foundation for the UAV's flight control system in this study. However, given that the tilt tri-rotor UAV's dynamic model is significantly more complex than conventional multirotor or fixed-wing aircraft, directly employing the standard Pixhawk 4 control architecture would be insufficient to achieve desired control performance. To address this challenge and leverage the maturity and stability advantages of commercial open-source flight controllers, we have:

- Maintained the original cascaded control framework while preserving the navigation layer.
- Modified the position and attitude loop controllers to accommodate tilt tri-rotor dynamics.
- Redesigned both the strategy layer and the control allocator layer.

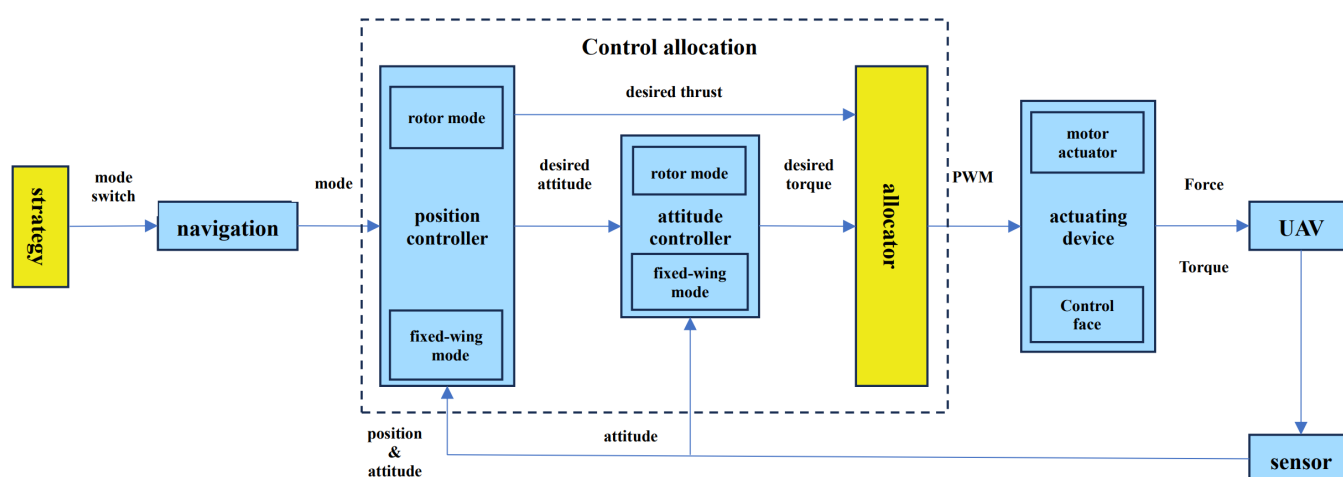


Figure 3. Flight control architecture.

3. Dynamic Model Establishment of Tilt-Rotor Tricopter

3.1. Dynamic Model

According to Newton's second law, by combining the kinematic relationships between angular velocity and attitude angles, the rigid body dynamic model of the UAV in the body coordinate system can be established as the following nine-dimensional equations:

$$\begin{cases} \begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = \begin{bmatrix} rv - qw \\ pw - ru \\ qu - pv \end{bmatrix} + \frac{1}{m} \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} \\ \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} \Gamma_1 pq - \Gamma_2 qr \\ \Gamma_5 pr - \Gamma_6 (p^2 - r^2) \\ \Gamma_7 pq - \Gamma_1 qr \end{bmatrix} + \begin{bmatrix} \Gamma_3 M_x + \Gamma_4 M_z \\ 1/J_y M_y \\ \Gamma_4 M_x + \Gamma_8 M_z \end{bmatrix} \\ \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \end{cases} \quad (1)$$

where $\mathbf{F} = [F_x, F_y, F_z]^T$ represents the resultant external force acting on the UAV in the body frame, m denotes the takeoff mass of the UAV, $\mathbf{M} = [M_x, M_y, M_z]^T$ is the resultant external moment about the center of mass expressed in the body frame, $\mathbf{V} = [u, v, w]^T$ indicates the translational velocity of the UAV in the body frame, $\boldsymbol{\omega} = [p, q, r]^T$ is the angular velocity of the UAV in the body frame, J_x, J_y, J_z each represent the moment of inertia about x-axis, y-axis, and z-axis, respectively, ϕ, θ, ψ each represents the roll, pitch, and yaw attitudes, respectively, and $\Gamma = J_x J_z - J_{xz}^2$, $\Gamma_1 = \frac{J_{xy}(J_x - J_y + J_z)}{\Gamma}$, $\Gamma_2 = \frac{J_z(J_z - J_y) + J_{xz}^2}{\Gamma}$, $\Gamma_3 = \frac{J_z}{\Gamma}$, $\Gamma_4 = \frac{J_{xz}}{\Gamma}$, $\Gamma_5 = \frac{(J_z - J_x)}{J_y}$, $\Gamma_6 = \frac{J_{xz}}{J_y}$, $\Gamma_7 = \frac{J_x(J_x - J_y) + J_{xz}^2}{\Gamma}$, and $\Gamma_8 = \frac{J_x}{\Gamma}$.

3.2. Model Simplification

To enhance the accuracy of aerodynamic simulation for rotary-wing UAV flight while reducing modeling complexity, the following assumptions are introduced:

- (1) The UAV is treated as a rigid body, and elastic deformation of the airframe is neglected;
- (2) The mass and mass distribution of the UAV are fixed and do not change with time;
- (3) The effects of the Earth's revolution and rotation are not considered;
- (4) Given the flight altitude and low-speed characteristics, the air is regarded as inviscid and incompressible;
- (5) The flapping and lead-lag motions of the rotor blades are not considered;
- (6) The anti-rotation gyroscopic torque caused by rotor tilting is neglected;
- (7) The aerodynamic interactions between rotors, as well as between wings and rotors, are ignored;
- (8) The tilt-rotor UAV is bilaterally symmetric with respect to the airframe axis system. The products of inertia J_{xy} , J_{xz} , and J_{yz} are relatively small compared to the moments of inertia J_x , J_y , and J_z about the x-, y-, and z-axes, and are therefore neglected.

Due to the significant impact of model complexity on simulation analysis, the model needs to be simplified for the control process of the tilt-rotor tricopter. By referring to other current research literature [24] and neglecting gyroscopic effects during flight, the nine-dimensional dynamic equation is simplified into a three-channel dynamic model considering roll angle, pitch angle, and yaw angle, as well as position, to accurately describe the dynamic characteristics of the tilt-rotor tricopter while reducing the complexity of the original model. Therefore, the dynamic model of the tilt-rotor tricopter can be transformed as follows:

$$\begin{aligned}
\ddot{x} &= \frac{u_1}{m} (\cos \phi \sin \theta \cos \varphi + \sin \phi \sin \varphi) - \frac{G_1 \dot{x}}{m} \\
\ddot{y} &= \frac{u_1}{m} (\cos \phi \sin \theta \sin \varphi - \sin \phi \cos \varphi) - \frac{G_2 \dot{y}}{m} \\
\ddot{z} &= \frac{u_1}{m} (\cos \phi \cos \theta) - g - \frac{G_3 \dot{z}}{m} \\
\ddot{\phi} &= \frac{a}{J_x} u_2 - \frac{G_4 a}{J_x} \dot{\phi} \\
\ddot{\theta} &= \frac{a}{J_y} u_3 - \frac{G_5 a}{J_y} \dot{\theta} \\
\ddot{\varphi} &= \frac{a}{J_z} u_4 - \frac{G_6 a}{J_z} \dot{\varphi}
\end{aligned} \tag{2}$$

where ϕ , θ and φ represent the roll angle, pitch angle, and yaw angle of the tilt-rotor tricopter, respectively. x , y and z represent the position of the tilt-rotor tricopter in three-dimensional space. m is the mass of the aircraft. g is the gravitational acceleration. a is the ratio of the anti-torque coefficient to the lift coefficient of the rotor under aerodynamic effects. J_x , J_y and J_z are the moments of inertia of the aircraft relative to the body coordinate system. G_i ($i = 1, 2, \dots, 6$) is the aerodynamic coefficient of the tilt-rotor tricopter, which consists of the drag coefficient C_D , side force coefficient C_Y , lift coefficient C_L , rolling moment coefficient C_l , pitching moment coefficient C_m , and yawing moment coefficient C_n , respectively, and $\mathbf{U} = [u_1 \ u_2 \ u_3 \ u_4]$ is the control input.

Remark 1. The model simplification strategy employed in this study adheres to the control-oriented modeling principle, aiming to capture the dominant dynamic characteristics of the tilt-rotor tricopter while neglecting secondary coupling effects to streamline controller design. To address the resulting model uncertainties, the proposed control architecture incorporates a Linear Extended State Observer (LESO) to actively estimate and compensate for unmodeled dynamics and external disturbances, thereby ensuring robustness. Furthermore, hardware-in-the-loop (HIL) simulations and comparative analyses with established studies confirm that the simplified model effectively captures the essential system dynamics, and the designed controller demonstrates satisfactory performance in trajectory tracking and attitude stabilization in practical applications.

3.3. Model Transformation

The tilt-rotor tricopter is a strongly coupled system. To simplify the controller, this paper combines the dual-loop control idea and divides the control model into position control and attitude control. Considering the strong internal coupling and strong external interference during the flight of the tilt-rotor tricopter, the dynamic model is transformed into the following attitude subsystem and position subsystem.

3.4. Attitude Subsystem

$$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = a_1 x_2 + b_1 u_2 + d_1. \\ \dot{x}_3 = x_4, \\ \dot{x}_4 = a_2 x_4 + b_2 u_3 + d_2. \\ \dot{x}_5 = x_6, \\ \dot{x}_6 = a_3 x_6 + b_3 u_4 + d_3. \end{cases} \tag{3}$$

where the subsystem parameters are $a_1 = -\frac{G_4 a}{J_x}$, $a_2 = -\frac{G_5 a}{J_y}$, and $a_3 = -\frac{G_6 a}{J_z}$, $b_1 = \frac{a}{J_x}$, $b_2 = \frac{a}{J_y}$, and $b_3 = \frac{a}{J_z}$, $[x_1, x_2, x_3, x_4, x_5, x_6]^T = [\phi, \dot{\phi}, \theta, \dot{\theta}, \varphi, \dot{\varphi}]^T$ represents the state variables of the attitude subsystem, and d_i ($i = 1, 2, 3$) represents the disturbance received by the attitude subsystem of the aircraft.

3.5. Position Subsystem

$$\begin{cases} \dot{x}_7 = x_8, \\ \dot{x}_8 = a_4 x_8 + \frac{\cos x_1 \cos x_3}{m} u_1 - g + d_4, \\ \dot{x}_9 = x_{10}, \\ \dot{x}_{10} = a_5 x_{10} + \frac{u_x}{m} u_1 + d_5, \\ \dot{x}_{11} = x_{12}, \\ \dot{x}_{12} = a_6 x_{12} + \frac{u_y}{m} u_1 + d_6. \end{cases} \quad (4)$$

where the position subsystem parameters are $a_4 = -\frac{G_3}{m}$, $a_5 = -\frac{G_1}{m}$, and $a_6 = -\frac{G_2}{m}$, $[x_7, x_8, x_9, x_{10}, x_{11}, x_{12}]^T = [z, \dot{z}, x, \dot{x}, y, \dot{y}]^T$ represents the state variables of the position subsystem, u_x, u_y are the control inputs of the x-position subsystem and y-position subsystem, respectively, and $d_i (i = 4, 5, 6)$ represents the disturbance received by the position subsystem of the aircraft.

3.6. Assumptions

To facilitate controller design and stability analysis, the following commonly used assumptions are made, which are widely adopted in existing literature on rotor/fixed-wing aircraft control systems.

Assumption 1. The given three-dimensional reference signals ϕ_d, θ_d and φ_d are continuously differentiable.

Assumption 2. There exists a constant ξ such that the external disturbance $d_i (i = 1, 2, \dots, 6)$ satisfies $|d_i| \leq \xi_i$.

Assumption 3. For the filtering output of the virtual control input α_{id} , there exists a normal constant r_{id} such that $|\alpha_{id}| \leq r_{id}$ always holds.

The above assumptions are widely adopted in the existing literature on rotor/fixed-wing aircraft control systems. Therefore, this paper directly incorporates these assumptions into the controller design process without additional proof. Based on these assumptions, this paper introduces a barrier Lyapunov function to limit the aircraft state during the controller design phase, uses a disturbance observer to offset internal and external disturbances, and adopts a filter to estimate the derivative of the virtual control law to address the common “exponential explosion” problem in backstepping control.

4. Backstepping Controller Design with State Constraints

This section presents the designs of backstepping controllers for different subsystems using barrier Lyapunov functions. While ensuring that the position and attitude errors are bounded, disturbance observers are used to estimate internal and external disturbances to ensure the stability of the tilt-rotor tricopter during the transition mode.

4.1. Attitude Controller Design

The attitude subsystem can be decomposed into roll angle subsystem, pitch angle subsystem, and yaw angle subsystem control during the controller design process.

4.1.1. Roll Angle Subsystem

First, consider the roll angle subsystem given by Equation (5).

$$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = a_1 x_2 + b_1 u_2 + d_1. \end{cases} \quad (5)$$

Step 1: Design of the virtual input x_1^d

The tracking error of x_1 is defined as

$$e_1 = x_1 - \phi_d \quad (6)$$

Taking the derivative of the error term yields

$$\dot{e}_1 = \dot{x}_1 - \dot{\phi}_d = e_2 + x_1^d - \dot{\phi}_d \quad (7)$$

where e_2 represents the error in x_2 tracking the virtual control law x_1^d , defined as

$$e_2 = x_2 - x_1^d \quad (8)$$

To ensure that the tilt-rotor UAV's state variables satisfy predefined state constraints, a barrier Lyapunov function (BLF) is designed in the following form:

$$L_1 = \frac{k_{b1}^2}{\pi} \tan\left(\frac{\pi e_1}{2k_{b1}}\right) \quad (9)$$

Combining the error derivative expression in Equation (7), the time derivative of this function is obtained as

$$\dot{L}_1 = \frac{e_1}{\cos^2\left(\frac{\pi e_1}{2k_{b1}}\right)} (e_2 + x_1^d - \dot{\phi}_d) \quad (10)$$

Based on the derivative result, the virtual control input is designed as

$$x_1^d = \dot{\phi}_d - c_1 e_1 \quad (11)$$

Substituting the designed virtual control input Equation (11) back into the time derivative of the Lyapunov function Equation (10) yields

$$\dot{L}_1 = -c_1 \frac{e_1^2}{\cos^2\left(\frac{\pi e_1}{2k_{b1}}\right)} + \frac{e_1 e_2}{\cos^2\left(\frac{\pi e_1}{2k_{b1}}\right)} \quad (12)$$

Step 2: Design of the control input

Based on the second-order error dynamics in Equation (8), its time derivative is derived as

$$\dot{e}_2 = a_1 x_2 + b_1 u_2 + d_1 - \dot{x}_1^d \quad (13)$$

To address the unknown disturbance d_1 in the equation, a disturbance observer is designed as follows:

$$\begin{cases} \hat{d}_1 = p_1 + l_1 e_2, \\ \dot{p}_1 = -l_1 (a_1 x_2 + b_1 u_2 + \hat{d}_1 - \dot{x}_1^d) \end{cases} \quad (14)$$

where the estimation error of the disturbance d_1 is given by

$$\tilde{d}_1 = d_1 - \hat{d}_1 \quad (15)$$

Combining Equation (15) with the derivative of the estimated term, the time derivative of the estimation error is expressed as

$$\dot{\tilde{d}}_1 = \dot{d}_1 - l_1 \tilde{d}_1 \quad (16)$$

Here, l_1 is a design parameter of the disturbance observer, tuned based on observation results. p_1 is the internal observer state.

To circumvent the difficulty in analytically deriving the virtual control law, a first-order filter is employed to estimate its derivative:

$$\dot{x}_1^c = \frac{-(x_1^c - x_1^d)}{\tau_1} \quad (17)$$

where τ_1 is a positive constant, x_1^c denotes the estimate of x_1^d , and the filtering estimation error is defined as

$$e_{1d} = \dot{x}_1^c - \dot{x}_1^d \quad (18)$$

For the second-order roll angle subsystem, a quadratic Barrier Lyapunov Function (BLF) is selected:

$$L_2 = L_1 + \frac{1}{2}e_2^2 \quad (19)$$

Its time derivative yields

$$\dot{L}_2 = e_2(a_1x_2 + b_1u_2 + d_1 - \dot{x}_1^d) + \dot{L}_1 \quad (20)$$

Using the derivative result and the expression of L_1 , the control input is constructed as

$$u_2 = \frac{1}{b_1} \left(-a_1x_2 - \hat{d}_1 + \dot{x}_1^c - c_2e_2 - \frac{e_1}{\cos^2\left(\frac{\pi e_1}{2k_{b1}^2}\right)} \right) \quad (21)$$

Substituting Equation (20) into Equation (19) leads to

$$\dot{L}_2 = -c_2e_2^2 - c_1 \frac{e_1^2}{\cos^2\left(\frac{\pi e_1}{2k_{b1}^2}\right)} + e_2\tilde{d}_1 + e_2e_{1d} - l_1\tilde{d}_1^2 + \tilde{d}_1\dot{d}_1 \quad (22)$$

Further, considering the estimation errors of both the observer and the filter, the composite Lyapunov function is chosen as

$$L_3 = L_2 + \frac{1}{2}\tilde{d}_1^2 + \frac{1}{2}e_{1d}^2 \quad (23)$$

whose time derivative satisfies

$$\begin{aligned} \dot{L}_3 &= -c_1 \frac{e_1^2}{\cos^2\left(\frac{\pi e_1}{2k_{b1}^2}\right)} - c_2e_2^2 + e_2\tilde{d}_1 + \tilde{d}_1\dot{\tilde{d}}_1 + e_{1d}\dot{e}_{1d} \leq \\ &-c_1 \frac{e_1^2}{\cos^2\left(\frac{\pi e_1}{2k_{b1}^2}\right)} - \left(c_2 - \frac{1}{2}\right)e_2^2 - (l_1 - 1)\tilde{d}_1^2 \\ &- \left(\frac{1}{2\tau_1} - \frac{1}{4}\right)e_{1d}^2 + \frac{1}{2}\zeta_1^2 + \frac{1}{4}r_{1d}^2 \leq -\rho_1 L_3 + M_1 \end{aligned} \quad (24)$$

where $\rho_1 = \min\left\{c_1, c_2 - \frac{1}{2}, l_1 - 1, \frac{1}{2\tau_1} - \frac{1}{4}\right\}$, $M_1 \geq \frac{1}{2}\zeta_1^2 + \frac{1}{4}r_{1d}^2$.

According to Lyapunov stability theory, the roll angle tracking error guarantees the following state constraint:

$$e_1 < k_{b1} \sqrt{1 - e^{-2(L_3(0) - \frac{M_1}{\rho_1})e^{-\rho_1 t} - 2\frac{M_1}{\rho_1}}} \leq k_{b1} \quad (25)$$

This confirms that the roll angle tracking error will persistently remain within the predefined bounds.

4.1.2. Pitch Angle Subsystem

Similarly, for the pitch angle subsystem given by Equation (26), the above process is repeated:

$$\begin{cases} \dot{x}_3 = x_4, \\ \dot{x}_4 = a_2x_4 + b_2u_3 + d_2. \end{cases} \quad (26)$$

Note that there exists an unknown disturbance d_2 in the equation, and a disturbance observer of the following form is designed to estimate it:

$$\begin{cases} \dot{\hat{d}}_2 = p_2 + l_2e_4, \\ \dot{p}_2 = -l_2(a_2x_4 + b_2u_3 + \hat{d}_2 - \dot{x}_3^d) \end{cases} \quad (27)$$

where l_2 is the design parameter of the disturbance observer. p_2 is the internal observer state.

The virtual control input is designed as

$$x_3^d = \dot{\theta}_d - c_3e_3 \quad (28)$$

The control input is constructed as

$$u_3 = \frac{1}{b_2} \left(-a_2x_4 - \hat{d}_2 + \dot{x}_3^d - c_4e_4 - \frac{e_3}{\cos^2\left(\frac{\pi e_3}{2k_{b3}^2}\right)} \right) \quad (29)$$

where $e_3 = x_3 - \theta_d$, $e_4 = x_4 - x_3^d$.

To ensure that the state variables of the tilt-rotor tricopter satisfy the predefined state constraints, a barrier Lyapunov function of the following form is designed for the pitch angle subsystem:

$$L_6 = L_5 + \frac{1}{2}\tilde{d}_2^2 + \frac{1}{2}e_{2d}^2 \quad (30)$$

where $L_5 = L_4 + \frac{1}{2}e_4^2$, $L_4 = \frac{k_{b3}^2}{\pi} \tan\left(\frac{\pi e_3}{2k_{b3}^2}\right)$, $\tilde{d}_2 = d_2 - \hat{d}_2$.

Taking the derivative of L_6 with respect to time yields

$$\begin{aligned} \dot{L}_6 &= -c_3 \frac{e_3^2}{\cos^2\left(\frac{\pi e_3}{2k_{b3}^2}\right)} - c_3e_4^2 + e_4\tilde{d}_2 + \tilde{d}_2\dot{\tilde{d}}_2 + e_{2d}\dot{e}_{2d} \\ &\leq -c_3 \frac{e_3^2}{\cos^2\left(\frac{\pi e_3}{2k_{b3}^2}\right)} - \left(c_4 - \frac{1}{2}\right)e_4^2 - (l_2 - 1)\tilde{d}_2^2 \\ &\quad - \left(\frac{1}{2\tau_2} - \frac{1}{4}\right)e_{2d}^2 + \frac{1}{2}\zeta_2^2 + \frac{1}{4}r_{2d}^2 \leq -\rho_2L_6 + M_2 \end{aligned} \quad (31)$$

where $\rho_2 = \min\left\{c_3, c_4 - \frac{1}{2}, l_2 - 1, \frac{1}{2\tau_2} - \frac{1}{4}\right\}$, $M_2 \geq \frac{1}{2}\zeta_2^2 + \frac{1}{4}r_{2d}^2$.

According to Lyapunov theory, the pitch angle error satisfies the following angular constraint:

$$e_3 < k_{b3} \sqrt{1 - e^{-2(L_6(0) - \frac{M_2}{\rho_2})e^{-\rho_2 t} - 2\frac{M_2}{\rho_2}}} \leq k_{b3} \quad (32)$$

4.1.3. Yaw Angle Subsystem

Similarly, for the yaw angle subsystem given by Equation (33), the above process is repeated:

$$\begin{cases} \dot{x}_5 = x_6, \\ \dot{x}_6 = a_3 x_6 + b_3 u_4 + d_3. \end{cases} \quad (33)$$

The disturbance observer is designed as

$$\begin{cases} \hat{d}_3 = p_3 + l_3 e_6, \\ \dot{p}_3 = -l_3 (a_3 x_6 + b_3 u_4 + \hat{d}_3 - \dot{x}_5^d) \end{cases} \quad (34)$$

The virtual control input is designed as

$$x_5^d = \dot{\psi}_d - c_5 e_5 \quad (35)$$

The control input is constructed as

$$u_4 = \frac{1}{b_3} \left(-a_3 x_6 - \hat{d}_3 + \dot{x}_5^d - c_6 e_6 - \frac{e_5}{\cos^2 \left(\frac{\pi e_5}{2k_{b5}^2} \right)} \right) \quad (36)$$

where $e_5 = x_5 - \psi_d$, $e_6 = x_6 - \dot{x}_5^d$, l_3 serves as the design parameter for the disturbance observer, and p_3 is the internal observer state.

Similar to the roll and pitch angle subsystems, the following BLF is designed:

$$L_9 = L_8 + \frac{1}{2} \tilde{d}_3^2 \quad (37)$$

where $L_8 = L_7 + \frac{1}{2} e_6^2$, $L_7 = \frac{k_{b5}^2}{\pi} \tan \left(\frac{\pi e_5}{2k_{b5}^2} \right)$, $\tilde{d}_3 = d_3 - \hat{d}_3$.

Taking the derivative of L_9 and applying Young's inequality yields

$$\begin{aligned} \dot{L}_9 &\leq -c_5 \frac{e_5^2}{\cos^2 \left(\frac{\pi e_5}{2k_{b5}^2} \right)} - \left(c_6 - \frac{1}{2} \right) e_6^2 - (l_3 - 1) \tilde{d}_3^2 \frac{1}{2} \zeta_3^2 \\ &\leq -\rho_3 L_9 + M_3 \end{aligned} \quad (38)$$

where $\rho_3 = \min \left\{ c_5, c_6 - \frac{1}{2}, l_3 - 1 \right\}$, $M_3 \geq \frac{1}{2} \zeta_3^2$.

According to Lyapunov theory, the yaw angle error satisfies an angular constraint of the form

$$e_5 < k_{b5} \sqrt{1 - e^{-2(L_9(0) - \frac{M_3}{\rho_3})e^{-\rho_3 t} - 2\frac{M_3}{\rho_3}}} \leq k_{b5} \quad (39)$$

Therefore, it can be concluded that the tracking errors of the yaw, pitch, and roll angles will remain within the preset boundaries.

4.2. Position Controller Design

The position subsystem can be divided into altitude subsystem, x position subsystem, and y position subsystem control during the controller design process.

4.2.1. Altitude Subsystem

First, consider the altitude subsystem given by Equation (40):

$$\begin{cases} \dot{x}_7 = x_8, \\ \dot{x}_8 = a_4 x_8 + \frac{\cos x_1 \cos x_3}{m} u_1 - g + d_4. \end{cases} \quad (40)$$

Step 1: Design of the virtual input x_7^d

The tracking error of x_7 is defined as

$$e_7 = x_7 - z_d \quad (41)$$

Taking the derivative of the error term yields

$$\dot{e}_7 = \dot{x}_7 - \dot{z}_d = e_8 + x_7^d - \dot{z}_d \quad (42)$$

where e_8 represents the error in x_8 tracking the virtual control law x_7^d , defined as

$$e_8 = x_8 - x_7^d \quad (43)$$

To ensure that the tilt-rotor UAV's state variables satisfy predefined state constraints, a barrier Lyapunov function (BLF) is designed in the following form:

$$L_{10} = \frac{k_{b7}^2}{\pi} \tan\left(\frac{\pi e_7}{2k_{b7}}\right) \quad (44)$$

Combining the error derivative expression Equation (42), the time derivative of this function is obtained as

$$\dot{L}_{10} = \frac{e_7}{\cos^2\left(\frac{\pi e_7}{2k_{b7}}\right)} (e_8 + x_7^d - \dot{z}_d) \quad (45)$$

Based on the derivative result, the virtual control input is designed as

$$x_7^d = \dot{z}_d - c_7 e_7 \quad (46)$$

Substituting the designed virtual control input Equation (46) back into the time derivative of the Lyapunov function in Equation (45) yields

$$\dot{L}_{10} = -c_7 \frac{e_7^2}{\cos^2\left(\frac{\pi e_7}{2k_{b7}}\right)} + \frac{e_7 e_8}{\cos^2\left(\frac{\pi e_7}{2k_{b7}}\right)} \quad (47)$$

Step 2: Design of the control input u_1

Based on the second-order error dynamics in Equation (43), its time derivative is derived as

$$\dot{e}_8 = a_4 x_8 + \frac{\cos x_1 \cos x_3}{m} u_1 - g + d_4 - \dot{x}_7^d \quad (48)$$

To address the unknown disturbance d_4 in the equation, a disturbance observer is designed as follows:

$$\begin{cases} \hat{d}_4 = p_4 + l_4 e_8, \\ \dot{p}_4 = -l_4 (a_4 x_8 + \frac{\cos x_1 \cos x_3}{m} u_1 - g + \hat{d}_4 - \dot{x}_7^d) \end{cases} \quad (49)$$

where the estimation error of the disturbance d_4 is given by

$$\tilde{d}_4 = d_4 - \hat{d}_4 \quad (50)$$

Combining Equation (50) with the derivative of the estimated term, the time derivative of the estimation error is expressed as

$$\dot{\tilde{d}}_4 = \dot{d}_4 - l_4 \tilde{d}_4 \quad (51)$$

Here, l_4 is a design parameter of the disturbance observer, tuned based on observation results, and p_4 is the internal observer state.

To circumvent the difficulty in analytically deriving the virtual control law, a first-order filter is employed to estimate its derivative:

$$\dot{x}_7^c = \frac{-(x_7^c - x_7^d)}{\tau_4} \quad (52)$$

where τ_4 is a positive constant, x_7^c denotes the estimate of x_7^d , and the filtering estimation error is defined as

$$e_{4d} = \dot{x}_7^c - \dot{x}_7^d \quad (53)$$

For the second-order altitude subsystem, a quadratic Barrier Lyapunov Function (BLF) is selected:

$$L_{11} = L_{10} + \frac{1}{2}e_8^2 \quad (54)$$

Its time derivative yields

$$\dot{L}_{11} = e_8 \left(a_4 x_8 + \frac{\cos x_1 \cos x_3}{m} u_1 - g + d_4 - \dot{x}_7^d \right) + \dot{L}_{10} \quad (55)$$

Using the derivative result and the expression of L_{10} , the control input is constructed as

$$u_1 = \frac{m}{\cos x_1 \cos x_3} \left(-a_4 x_8 - \hat{d}_4 + \dot{x}_7^c - c_8 e_8 + g - \frac{e_7}{\cos^2 \left(\frac{\pi e_7}{2k_{b7}^2} \right)} \right) \quad (56)$$

Substituting Equation (55) into Equation (54) leads to

$$\dot{L}_{11} = -c_8 e_8^2 - c_7 \frac{e_7^2}{\cos^2 \left(\frac{\pi e_7}{2k_{b7}^2} \right)} + e_8 \tilde{d}_4 + e_8 e_{4d} - l_4 \tilde{d}_4^2 + \tilde{d}_4 \dot{\tilde{d}}_4 \quad (57)$$

Further, considering the estimation errors of both the observer and filter, the composite Lyapunov function is chosen as

$$L_{12} = L_{11} + \frac{1}{2}\tilde{d}_4^2 \quad (58)$$

whose time derivative satisfies

$$\begin{aligned} \dot{L}_{12} &= -c_7 \frac{e_7^2}{\cos^2 \left(\frac{\pi e_7}{2k_{b7}^2} \right)} - c_8 e_8^2 + e_8 \tilde{d}_4 + \tilde{d}_4 \dot{\tilde{d}}_4 \leq \\ &-c_7 \frac{e_7^2}{\cos^2 \left(\frac{\pi e_7}{2k_{b7}^2} \right)} - \left(c_8 - \frac{1}{2} \right) e_8^2 - (l_4 - 1) \tilde{d}_4^2 + \\ &\frac{1}{2} \tilde{\varsigma}_4^2 \leq -\rho_4 L_{12} + M_4, \end{aligned} \quad (59)$$

where $\rho_4 = \min \left\{ c_7, c_8 - \frac{1}{2}, l_4 - 1, \frac{1}{2\tau_4} - \frac{1}{4} \right\}$, and $M_4 \geq \frac{1}{2}\tilde{\varsigma}_4^2$.

According to Lyapunov stability theory, the altitude tracking error guarantees the following state constraint:

$$e_7 < k_{b7} \sqrt{1 - e^{-2(L_{12}(0) - \frac{M_4}{\rho_4})e^{-\rho_4 t} - 2\frac{M_4}{\rho_4}}} \leq k_{b7} \quad (60)$$

This confirms that the altitude tracking error will persistently remain within the predefined bounds.

4.2.2. X Position Subsystem

Similarly, for the x position subsystem given by Equation (61), the above process is repeated:

$$\begin{cases} \dot{x}_9 = x_{10}, \\ \dot{x}_{10} = a_5 x_{10} + \frac{u_x}{m} u_1 + d_5. \end{cases} \quad (61)$$

A disturbance observer is designed in the following form to estimate and compensate for d_5 :

$$\begin{cases} \hat{d}_5 = p_5 + l_5 z_{10}, \\ \dot{p}_5 = -l_5 (a_5 x_{10} + \frac{u_x}{m} u_1 + \hat{d}_5 - \dot{x}_9) \end{cases} \quad (62)$$

where l_5 is the parameter to be determined, and p_5 is the internal observer state.

The virtual control input is designed as

$$x_9^d = \dot{x}_d - c_9 e_9 \quad (63)$$

The control input is constructed as

$$u_x = \frac{m}{u_1} \left(-a_5 x_{10} - \hat{d}_5 + \dot{x}_9^d - c_{10} e_{10} - \frac{e_9}{\cos^2 \left(\frac{\pi e_9}{2k_{b9}^2} \right)} \right) \quad (64)$$

where $e_9 = x_9 - x_d$, and $e_{10} = x_{10} - x_9^d$.

For the x-position subsystem, the following BLF is selected:

$$L_{15} = L_{14} + \frac{1}{2} \tilde{d}_5^2 \quad (65)$$

where $L_{14} = L_{13} + \frac{1}{2} e_{10}^2$, $L_{13} = \frac{k_{b9}^2}{\pi} \tan \left(\frac{\pi e_9}{2k_{b9}} \right)$, and $\tilde{d}_5 = d_5 - \hat{d}_5$.

Differentiating L_{15} with respect to time and applying Young's inequality yields

$$\begin{aligned} \dot{L}_{15} &= -c_9 \frac{e_9^2}{\cos^2 \left(\frac{\pi e_9}{2k_{b9}} \right)} - c_{10} e_{10}^2 + e_{10} \tilde{d}_5 + \tilde{d}_5 \dot{\tilde{d}_5} \leq \\ &-c_9 \frac{e_9^2}{\cos^2 \left(\frac{\pi e_9}{2k_{b9}} \right)} - \left(c_{10} - \frac{1}{2} \right) e_{10}^2 - (l_5 - 1) \tilde{d}_5^2 + \\ &\frac{1}{2} \tilde{\zeta}_5^2 \leq -\rho_5 L_{15} + M_5 \end{aligned} \quad (66)$$

where $\rho_5 = \min \left\{ c_9, c_{10} - \frac{1}{2}, l_5 - 1 \right\}$, and $M_5 \geq \frac{1}{2} \tilde{\zeta}_5^2$.

According to Lyapunov theory, it can be concluded that the x position error satisfies a constraint of the form

$$e_9 < k_{b9} \sqrt{1 - e^{-2(L_{15}(0) - \frac{M_5}{\rho_5})e^{-\rho_5 t} - 2\frac{M_5}{\rho_5}}} \leq k_{b9} \quad (67)$$

4.2.3. Y Position Subsystem

Similarly, for the y position subsystem given by Equation (68), the above process is repeated:

$$\begin{cases} \dot{x}_{11} = x_{12}, \\ \dot{x}_{12} = a_6 x_{12} + \frac{u_y}{m} u_1 + d_6. \end{cases} \quad (68)$$

A disturbance observer is designed to estimate and compensate for d_6 :

$$\begin{cases} \hat{d}_6 = p_6 + l_6 e_{12}, \\ \dot{p}_6 = -l_6(a_6 x_{12} + \frac{u_y}{m} u_1 + \hat{d}_6 - \dot{x}_{11}^d) \end{cases} \quad (69)$$

where l_6 is the parameter to be determined, and p_6 is the internal observer state.

The virtual control input is designed as

$$x_{11}^d = \dot{y}_d - c_{11} e_{11} \quad (70)$$

The control input is constructed as

$$u_y = \frac{m}{u_1} \left(-a_6 x_{12} - \hat{d}_6 + \dot{x}_{11}^c - c_{12} e_{12} - \frac{e_{11}}{\cos^2\left(\frac{\pi e_{11}}{2k_{b11}^2}\right)} \right) \quad (71)$$

where $e_{11} = x_{11} - y_d$, and $e_{12} = x_{12} - x_{11}^d$.

The following BLF is selected:

$$L_{18} = L_{17} + \frac{1}{2} \tilde{d}_6^2 \quad (72)$$

where $L_{17} = L_{16} + \frac{1}{2} e_{12}^2$, $L_{16} = \frac{k_{b11}^2}{\pi} \tan\left(\frac{\pi e_{11}}{2k_{b11}}\right)$, and $\tilde{d}_6 = d_6 - \hat{d}_6$.

Analogous to the altitude and x-position subsystems, the time derivative of L_{18} is derived as

$$\begin{aligned} \dot{L}_{18} &\leq -c_{11} \frac{e_{11}^2}{\cos^2\left(\frac{\pi e_{11}}{2k_{b11}^2}\right)} - \left(c_{12} - \frac{1}{2}\right) e_{12}^2 - (l_6 - 1) \tilde{d}_6^2 \frac{1}{2} \zeta_6^2 \\ &\leq -\rho_6 L_{18} + M_6 \end{aligned} \quad (73)$$

where $\rho_6 = \min\left\{c_{11}, c_{12} - \frac{1}{2}, l_6 - 1\right\}$, and $M_6 \geq \frac{1}{2} \zeta_6^2$.

In accordance with Lyapunov theory, it can be concluded that the position error satisfies a constraint of the form

$$e_{11} < k_{b11} \sqrt{1 - e^{-2(L_{18}(0) - \frac{M_6}{\rho_6})e^{-\rho_6 t} - 2\frac{M_6}{\rho_6}}} \leq k_{b11} \quad (74)$$

Remark 2. The physical angular limits of the tilt mechanism are incorporated into the simulation model as hard constraints. For the transition-mode maneuvers studied in this paper (e.g., the circular trajectory tracking task), the optimized controller generates commands that operate well within these limits, as evidenced by the smooth and bounded control signals in the HIL results in Section 6.

Remark 3. The classical backstepping framework was chosen for its structural compatibility with our core objectives. Its recursive design aligns naturally with the cascade (position/attitude) dynamics of the tilt-rotor system and, crucially, provides a direct and rigorous way to enforce state constraints via Barrier Lyapunov Functions (BLFs). While methods like Sliding-Mode Backstepping or ADRC offer robustness, they introduce complexities (e.g., chattering risk, less straightforward constraint integration) that our architecture avoids. The proposed design incorporates a Linear Extended State Observer (LESO) to actively estimate and compensate for disturbances, effectively addressing the robustness requirement, while the Aquila Optimizer automates parameter tuning. Thus, the combination (Backstepping + BLF + LESO) forms a cohesive, tailored solution for the state-constrained transition control problem.

Remark 4. The tangent-type BLF $L_j = \frac{k_{bi}^2}{\pi} \tan\left(\frac{\pi e_i}{2k_{bi}}\right)$, $j = 1, 4, 7, 10, 13, 16$, and $i = 1, 3, 5, 7, 9, 11$ is adopted to enforce the state constraint $|e_i| < k_{bi}$. The tangent form is chosen for its stronger boundary enforcement, suitability for angular states, and numerical stability. The selection of k_{bi} is based on the tilt-rotor tricopter UAV's physical limits, control performance specifications, and safety considerations.

Remark 5. The observer gain l_i , $i = 1, 2, \dots, 6$ determines the LESO bandwidth: A larger l_i accelerates disturbance estimation but increases sensitivity to measurement noise, while a smaller l_i slows the response but improves noise robustness. In this work, all LESO gains are optimized together with the controller parameters using the improved Aquila Optimizer, ensuring an optimal balance between estimation speed and noise tolerance.

4.3. Stability Analysis

Based on the preceding developments, the following main result is presented with its corresponding proof.

Theorem 1. Consider the tilt-rotor tricopter system described by Equation (2), the attitude subsystem Equation (3), and the position subsystem Equation (4). Under Assumptions 1–3, and with the backstepping controllers Equation (21), Equation (29), and Equation (36) for the attitude loop and Equation (56), Equation (64), and Equation (71) for the position loop, and the LESO disturbance observers Equation (14), Equation (27), Equation (34), Equation (49), Equation (62), and Equation (69) employed for derivative estimation, the following properties hold for the overall closed-loop system:

- (1) All signals in the closed-loop system, including state-tracking errors, observer errors, and filter errors, are uniformly ultimately bounded.
- (2) The tracking errors for roll angle, pitch angle, yaw angle, altitude, x position, and y position remain strictly within their predefined bounds for all $t \geq t_0$, given proper initial conditions.

Proof. We select the following composite Lyapunov candidate for the closed-loop system:

$$L = L_3 + L_6 + L_9 + L_{12} + L_{15} + L_{18} \quad (75)$$

Taking the time derivative of L and substituting Equation (24), Equation (31), Equation (38), Equation (59), Equation (66), and Equation (73) yields

$$\dot{L} \leq -AL + B \quad (76)$$

where $A = \min\{\rho_1, \rho_2, \rho_3, \rho_4, \rho_5, \rho_6\}$, and $B = \sum_{i=1}^6 M_i$. This inequality proves that the entire closed-loop system is uniformly ultimately bounded.

It follows from Equation (25), Equation (32), Equation (39), Equation (60), Equation (67), and Equation (74) that the roll angle, pitch angle, yaw angle, altitude, x-position, and y-position obey their predefined bounds for all $t \geq t_0$, provided that the initial conditions are appropriate.

This completes the proof. $\square$

5. Controller Parameter Optimization Based on Improved Aquila Optimizer Algorithm

The proposed control architecture integrates multiple functional components, including a barrier Lyapunov function (BLF) for state constraints, a first-order filter to mitigate “complexity explosion” in backstepping control, and a linear extended state observer (LESO) for disturbance compensation. Each component involves multiple adjustable hyperparameters. Manual tuning, the traditional approach adopted in related studies [16–18], is not only time-consuming but also prone to suboptimal parameter combinations. An efficient metaheuristic optimization algorithm is therefore essential to automate and optimize the parameter tuning process. In addition, prior studies on tilt-rotor UAV control have rarely combined advanced optimization algorithms with state-constrained backstepping control. For example, sliding mode control (SMC) variants [20–23] face chattering issues and complex design, while adaptive control [6] relies heavily on parameter identification accuracy. These methods lack systematic parameter optimization, leading to compromised performance. Aquila Optimizer (AO)’s metaheuristic nature enables it to handle the high-dimensional, non-convex parameter space of the proposed controller without relying on precise model linearization or gradient information.

5.1. Algorithm Principle

The Aquila Optimizer (AO) is a metaheuristic algorithm inspired by the hunting behavior of Aquila, which incorporates four distinct optimization strategies to balance convergence speed and optimization capability at different iteration stages. In this study, an improved AO is employed to optimize controller parameters, aiming to achieve superior control performance. During the first two-thirds of the iterations, Method 1 and Method 2 are utilized to enhance global exploration, enabling rapid convergence of the optimization model. In the remaining one-third of the iterations, Method 3 and Method 4 are applied for local exploitation, facilitating precise refinement to locate the optimal solution. The iterative process is illustrated in Figure 4.

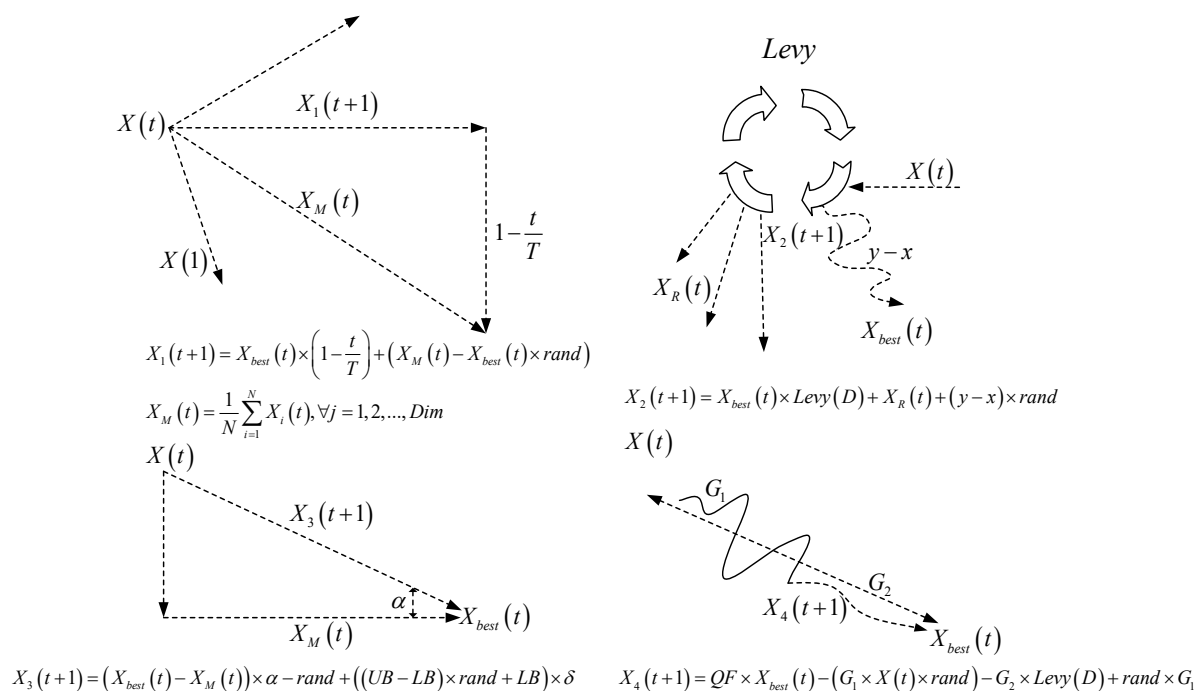


Figure 4. The iterative approach of the AO algorithm.

5.2. Improved Population Initialization Strategy

The traditional Aquila Optimizer (AO) only designs the iterative optimization process without considering population initialization techniques. This results in insufficient diversity and uniformity in the initial population distribution due to random initialization, thereby affecting the algorithm's performance, since the optimization capability of AO is influenced by the quality of the initial population. To address the population initialization process, this study introduces a chaotic elite opposition-based learning initialization strategy to mitigate the uncertainty caused by random initialization and enhance the algorithm's global optimization ability.

The initial population of the Aquila Optimizer (AO) is initialized using a chaotic map:

First, a vector $m \in [-1, 1]$ is generated and iteratively updated using the following equation to produce $N - 1$ new individuals. Subsequently, the individual vectors x_i of the AO are generated through mapping calculations. This method ensures that the initial population exhibits enhanced randomness and universality, enabling non-repetitive traversal of all states within a certain range:

$$\begin{aligned} 2m_{i+1} &= 4m_i^3 - 3m_i; \\ x_i &= X_{lb} + \frac{(X_{ub} - X_{lb})(m_i + 1)}{2} \end{aligned} \quad (77)$$

To further expand the selection range of the initial population and improve its quality, the opposition-based learning (OBL) strategy is employed. Let x_j and x'_j denote the initial vector of the AO and its opposition-based counterpart, respectively. These vectors are computed using the following formula:

$$x'_j = \frac{a_j + b_j}{2} + \frac{a_j + b_j}{2k} - \frac{x_j}{k}. \quad (78)$$

where a_j and b_j present the lower and upper bounds of the population generation, respectively, and k is the scaling coefficient in the opposition-based learning mapping.

The process of enhancing population initialization using the chaotic elite opposition-based learning strategy is outlined as follows:

- (1) Generate an initial Aquila population with individuals.
- (2) Transform the initial population into a chaotic population using a chaotic map.
- (3) Construct an opposition-based population from the initial population.
- (4) Evaluate the fitness of individuals in X , Y and Z , then select the best-performing candidates to form an elite chaotic opposition-based population $X' = [X'_{i1}, \dots, X'_{id}]^T$

By integrating this enhanced initialization strategy with the iterative Aquila Optimizer (AO) algorithm, an improved AO variant is developed for optimizing controller parameters. The overall iterative optimization process is illustrated in Figure 5.

The improved Aquila Optimizer, owing to the enhanced quality of its initial iterative population, can compensate for the optimization iteration speed and search capability of the original algorithm, thereby enabling faster convergence to superior controller parameters.

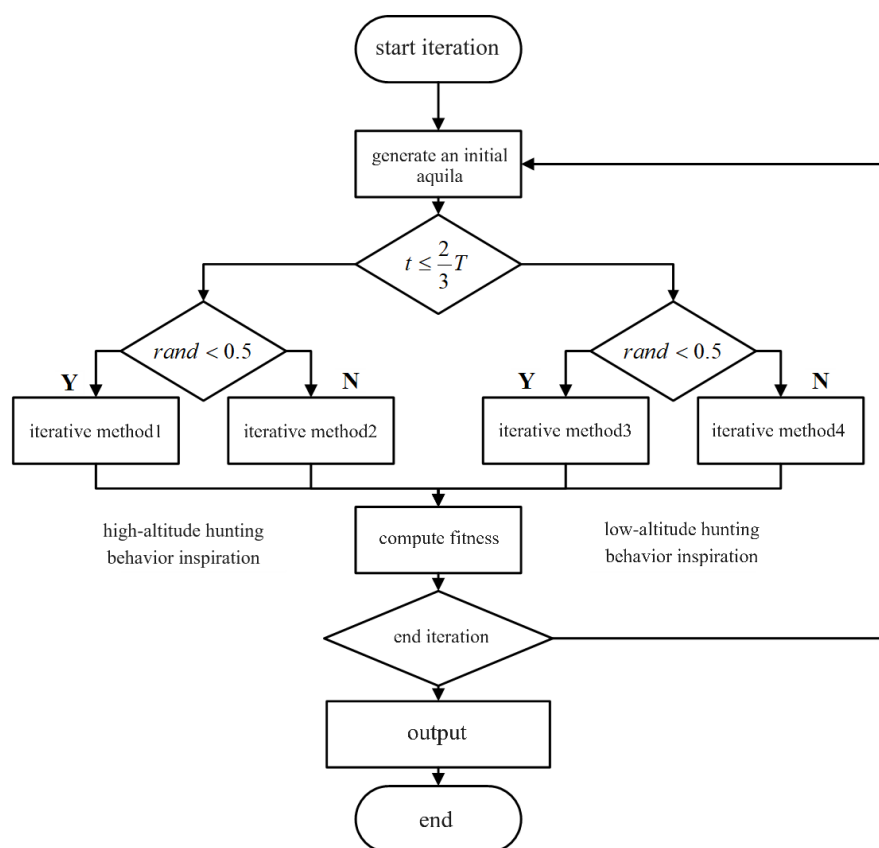


Figure 5. The process of the improved AO algorithm.

6. Simulation Analysis

A hardware-in-the-loop (HIL) simulation environment is constructed using Pixhawk4 (firmware version: px4fmu-v5) and MATLAB (2021b) Simulink, as is shown in Figure 6. Pixhawk Pilot Support Package (PSP) Toolbox is a Simulink toolbox officially released by Math works, which can employ the Embedded Coder in Simulink to automatically compile and deploy Simulink model autopilot algorithms to Pixhawk hardware systems. The UAV performs a trajectory tracking simulation of the transition from rotor mode to fixed-wing mode along a circular path with a radius of 5 m, a simulation step of 0.01s, and a simulation duration of 12 s. The parameters of the UAV dynamic model are shown in Table 2 and Figure 7.

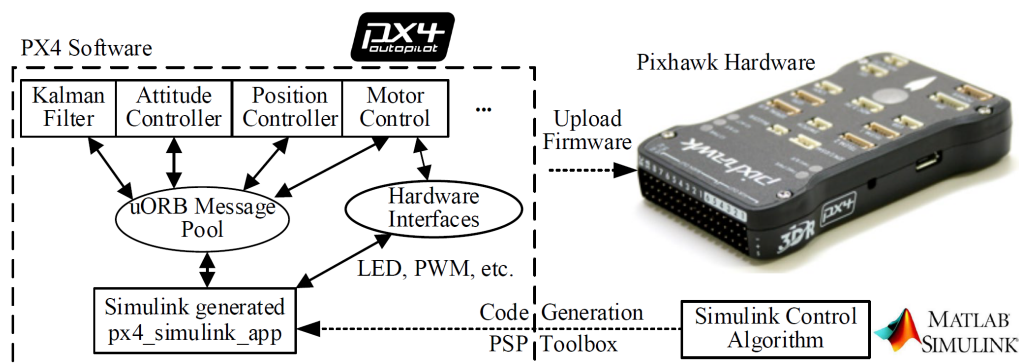
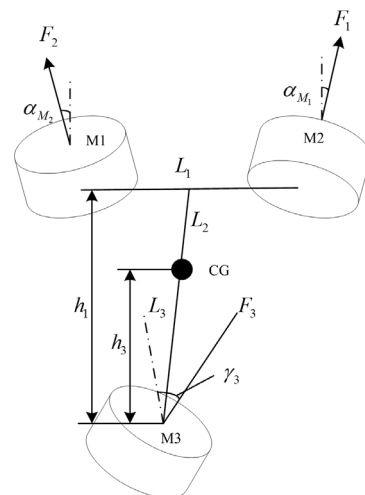


Figure 6. Hardware-in-the-loop (HIL) simulation environment based on Pixhawk4.

Table 2. Parameters of the UAV dynamic model.

Variable	Description	Value
m	UAV Mass	9 kg
g	Gravitational acceleration	9.8 m/s^2
L_1	Distance between M_1 and M_2 (front rotors) thrust application points	1000 mm
L_2	Distance from M_1 and M_2 to UAV Center of Gravity	350 mm
L_3	Distance from M_3 (rear rotor) to UAV Center of Gravity	532 mm
γ_3	Angle Between M_3 thrust line and body coordinate Z -axis	20°
I_{xx}	Moment of inertia about x-axis (rotor mode)	$2.1653 \text{ kg}\cdot\text{m}^2$
I_{yy}	Moment of inertia about y-axis (rotor mode)	$2.5376 \text{ kg}\cdot\text{m}^2$
I_{zz}	Moment of inertia about z-axis (rotor mode)	$0.6027 \text{ kg}\cdot\text{m}^2$
I_{xx}	Moment of inertia about x-axis (fixed-wing mode)	$2.1879 \text{ kg}\cdot\text{m}^2$
I_{yy}	Moment of inertia about y-axis (fixed-wing mode)	$2.7413 \text{ kg}\cdot\text{m}^2$
I_{zz}	Moment of inertia about z-axis (fixed-wing mode)	$0.6314 \text{ kg}\cdot\text{m}^2$
k_f	Rotor thrust coefficient	8.017×10^{-5}
k_m	Rotor anti-torque coefficient	1.6606×10^{-6}
T_{ms}	Time constant of the first-order inertial element of the servo	0.03 s
T_{mr}	Time constant of the first-order inertial element of the rotor	0.0167 s

**Figure 7.** Parameters of the UAV dynamic model.

Building upon the aforementioned HIL simulation environment, two simulation cases are designed to test the effectiveness and robustness of the proposed method. Specifically, in Case 1, the proposed method—referred to as the IAO-based backstepping (IAOBS) controller—is compared with the traditional backstepping (BS) controller and the PID controller to demonstrate its superiority. In Case 2, taking the Position system as an example, disturbances and noise are introduced to conduct a stress test on the proposed controller, thereby verifying the effectiveness of the designed disturbance observer and the robustness of the proposed method.

Figure 8 illustrates the convergence curve of the parameter optimization process, showing that IAO achieves rapid convergence. Table 3 presents the parameters before and after IAO optimization for the roll angle subsystem as an example. The optimized

parameters correspond to those used in the proposed method, while the pre-optimized parameters serve as the settings for the BS controller.

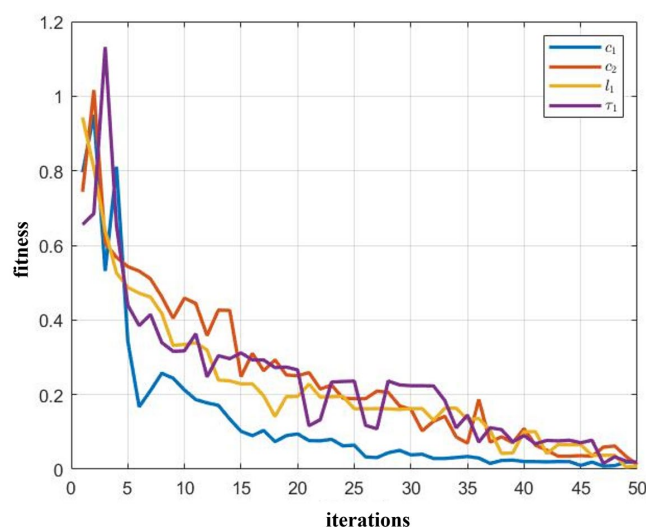


Figure 8. The fitness evolution curve of the controller parameters in the roll angle subsystem.

Table 3. Optimization results of the roll angle subsystem.

Parameter	Before IAO	After IAO
c_1	10	7.4261
c_2	5	4.2764
l_1	1	1.3285
τ_1	0.5	0.2265

6.1. Case 1

As shown in Figures 9–13, the simulation results for this case are presented. From the position and velocity tracking curves in Figures 9 and 10, it can be observed that compared with the BS and PID controllers, the Aquila-optimized IAOBS controller achieves improvements in both steady-state and transient performance in terms of position and velocity tracking. Particularly in the altitude direction, the optimized altitude subsystem controller successfully stabilizes the originally divergent system, ensuring convergence.

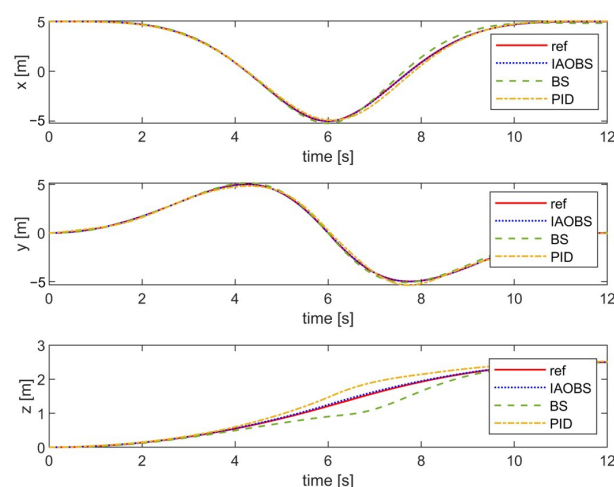


Figure 9. Position track curve.

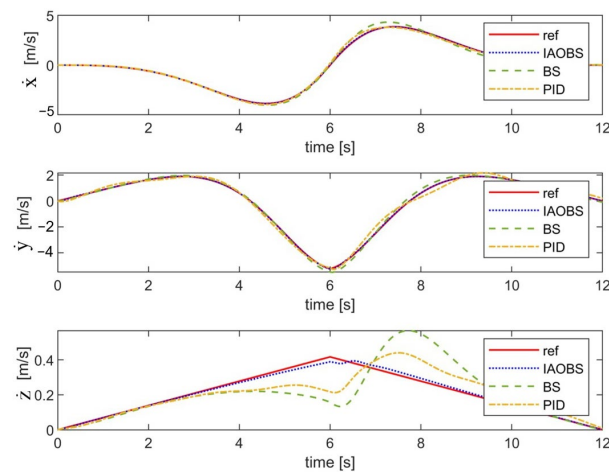


Figure 10. Velocity track curve.

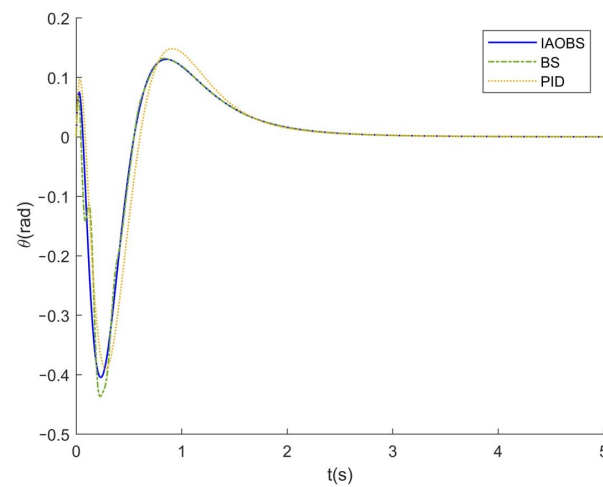


Figure 11. Pitch angle curve.

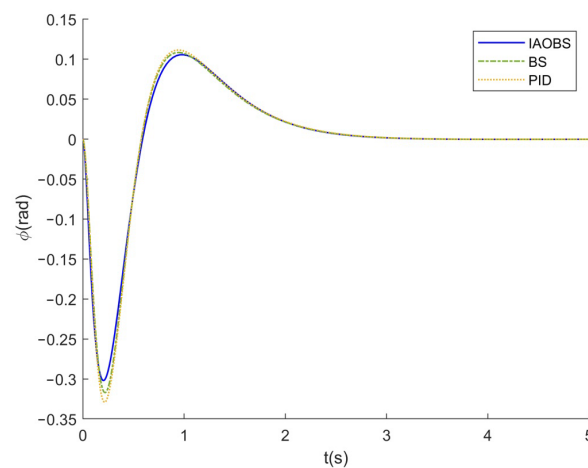


Figure 12. Roll angle curve.

Furthermore, the pitch, roll, and yaw angle curves shown in Figures 11–13 indicate that the proposed IAOBS controller yields smoother angle responses and exhibits better transient performance than the BS and PID controllers. In summary, by incorporating state constraints, the proposed method enhances the performance of the adaptive backstepping controller. This not only guarantees that the system error remains within predefined bounds but also significantly improves the response speed and convergence accuracy.

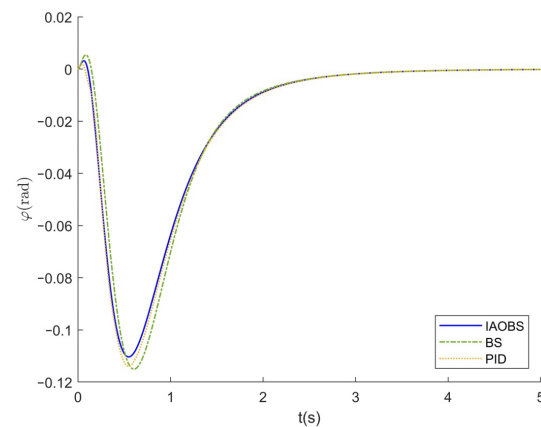


Figure 13. Yaw angle curve.

6.2. Case 2

To rigorously evaluate the robustness and disturbance rejection capability of the proposed IAOBS controller, a second simulation case was conducted exclusively on the Position system under a challenging composite scenario. The test environment was designed to emulate realistic flight conditions by introducing multiple sources of interference. Specifically, external disturbances, formulated as $d_4 = 0.5 \cos(t) + 0.5 \sin(1.5t)$, $d_5 = 0.4 \cos(t) + 0.6 \sin(2t)$, and $d_6 = 0.3 \cos(2t) + 0.7 \sin(t)$, were injected into the system dynamics to simulate the effect of wind gusts. Furthermore, to assess the controller's sensitivity to common implementation imperfections, Gaussian noise $n(t) \sim \mathcal{N}(0, 0.01)$ was added to the control inputs of all three subsystems, and the sensor measurements for altitude, x-position, and y-position were corrupted with both additive noise $n(t)$ and a constant measurement bias of 0.01.

The performance under these adverse conditions is comprehensively illustrated in Figures 14–18. The position and velocity tracking curves, presented in Figures 14 and 15, respectively, demonstrate that the IAOBS controller successfully maintains accurate trajectory following despite the significant disturbances and measurement corruption. The corresponding tracking errors, detailed in the position and velocity tracking error curves (Figures 16 and 17), remain bounded within small magnitudes, exhibiting no evident divergence or instability. This indicates that the state-constrained adaptive backstepping framework effectively compensates for the adverse effects, confining the error dynamics within the predefined performance boundaries.

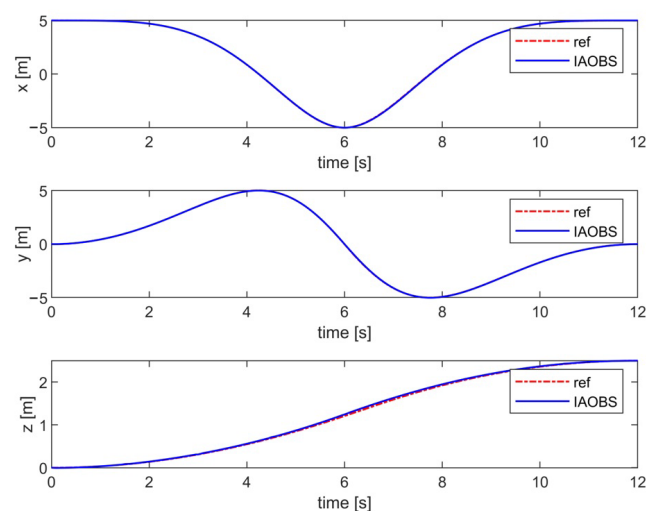


Figure 14. Position track curve.

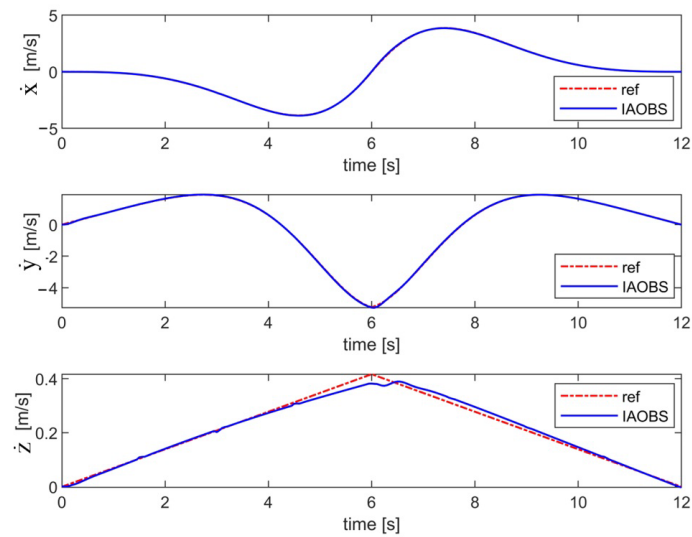


Figure 15. Velocity track curve.

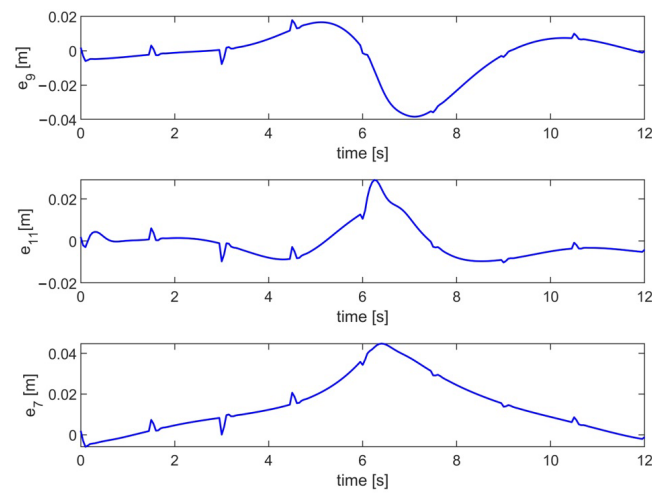


Figure 16. Position tracking error curve.

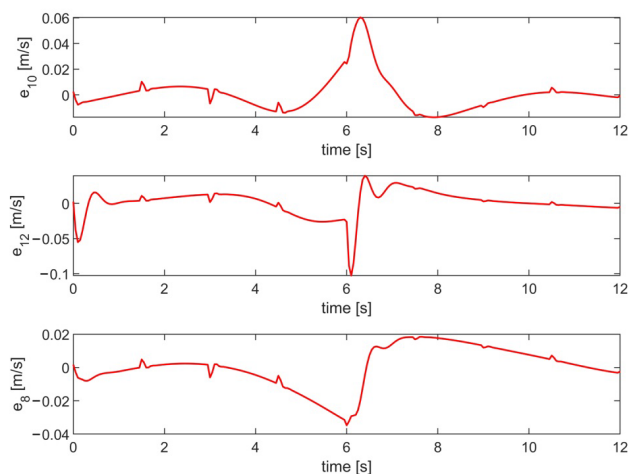


Figure 17. Velocity tracking error curve.

A key highlight of this case is the performance of the integrated disturbance observer, as shown in the disturbance estimation curve (Figure 18). This figure reveals a close match between the estimated and the actual injected disturbances. The observer accurately captures both the amplitude and phase characteristics of the complex, multi-frequency

disturbance signals in real-time. This precise estimation is the cornerstone of the controller's robustness, as it allows for effective feedforward compensation, thereby significantly mitigating the impact of disturbances on the tracking performance.

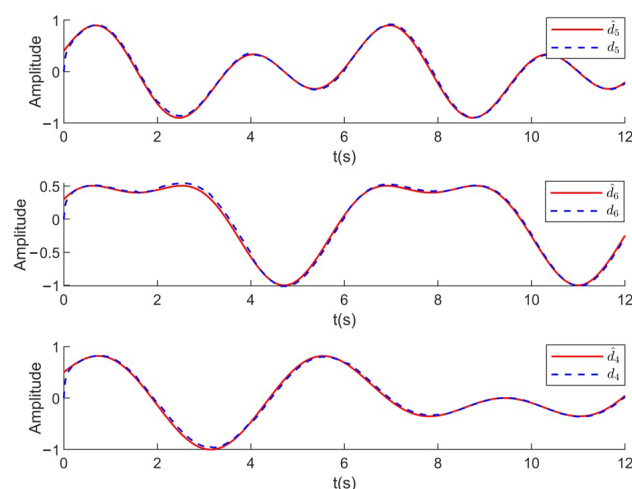


Figure 18. Disturbance estimation curve.

In summary, the results from Case 2 conclusively verify the robustness of the proposed IAOBS method. It maintains stable and precise tracking performance in the presence of compounded realistic challenges, including complex external disturbances, actuator noise, and sensor inaccuracies. The effectiveness of the disturbance observer is fully validated, confirming its critical role in enhancing the system's resilience and operational reliability under non-ideal conditions.

7. Conclusions

This paper addresses the anti-disturbance control problem during the transition mode of a tilt-rotor tricopter and proposes a backstepping controller design with state constraints.

Based on control-oriented modeling, a barrier Lyapunov function is introduced to ensure that the control variables remain within the allowable error range, a disturbance observer is used to offset disturbances during flight, and a filter is employed to handle the “differential explosion” problem.

The Aquila Optimizer (AO) algorithm is used to optimize the controller parameters. Simulation results show that the controller meets the pre-designed performance standards and improves the response of the control curves compared to the nominal controller, verifying the effectiveness of the proposed control method.

Regarding the limitations of the method proposed in this paper, there are two main points. First, the method exhibits model dependency—in scenarios involving extreme maneuvers or strong aerodynamic disturbances, these unmodeled dynamics may affect control accuracy. Second, while the simulation validated trajectory tracking performance of the transition from rotor mode to fixed-wing mode, a full-state, real-world physical experimental flight has yet to be verified.

For future work, several promising research directions can be explored to further enhance the performance, adaptability, and practicality of tilt-rotor tricopter control systems:

- (1) Current disturbance observers primarily address general internal/external perturbations but lack targeted modeling of wind field dynamics. Future work can incorporate advanced wind estimation techniques—such as extended Kalman filters (EKF) and unscented Kalman filters (UKF).

- (2) To reduce reliance on precise dynamic models and enhance adaptability to parameter variations (e.g., payload changes, rotor wear) or unmodeled nonlinearities, future controllers can fuse backstepping control with learning-based methods.
- (3) While the hardware-in-the-loop (HIL) simulation provides a reliable testbed, future research should extend to full-mode physical flight experiments—especially focusing on the entire transition process from rotor mode to fixed-wing mode.

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